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Inventor(s)	Chung; Chi-Wen et al.

Rotary bearing assembly

Abstract

A rotary bearing assembly includes an input shaft, an inner-ring component, an outer-ring component and a load element. The input shaft is configured to combine a rotating shaft of a motor to provide a power input. The inner-ring component includes a gear set, wherein the inner-ring component is sleeved on the input shaft through the gear set and driven by the input shaft. The outer-ring component is sleeved on the inner-ring component through a load element and engaged with the gear set, wherein when the gear set is driven by the input shaft to drive the inner-ring component, the gear set drives the outer-ring component, and the inner-ring component and the outer-ring component are rotated relatively, wherein one of the inner-ring component and the outer-ring component is served to provide a power output, and a rotational speed difference is between the power input and the power output.

Inventors: **Chung; Chi-Wen (Taoyuan, TW), Lin; Hung-Wei (Taoyuan, TW), Tsai; Hsien-Lung (Taoyuan, TW), Chu; Wei-Ying (Taoyuan, TW), Chen; Chin-Hsiang (Taoyuan, TW)**

Applicant: **Delta Electronics, Inc.** (Taoyuan, TW)

Family ID: **90943674**

Assignee: **Delta Electronics, Inc. (Taoyuan, TW)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0335944	12/2016	Nishimura	N/A	F16H 57/082
2018/0031079	12/2017	Yoshida	N/A	F16H 57/02004
2019/0162281	12/2018	Nakamura	N/A	B62D 5/04
2019/0285143	12/2018	Sasaki	N/A	F16H 57/023
2019/0331199	12/2018	Cao	N/A	F16H 1/32
2022/0082156	12/2021	Makisumi	N/A	N/A
2022/0136587	12/2021	Chung et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101769373	12/2009	CN	N/A
202418243	12/2011	CN	N/A
203516608	12/2013	CN	N/A
104121334	12/2013	CN	N/A
104565217	12/2014	CN	B25J 17/00
109915547	12/2018	CN	N/A
210106448	12/2019	CN	N/A
111120518	12/2019	CN	N/A
211175109	12/2019	CN	N/A
111828550	12/2019	CN	B25J 9/102
H06-037646	12/1993	JP	N/A
H11-210843	12/1998	JP	N/A
2006-200557	12/2005	JP	N/A
2007-046730	12/2006	JP	N/A
2008-089144	12/2007	JP	N/A
2010-032000	12/2009	JP	N/A

2021-067317	12/2020	JP	N/A
2021179232	12/2020	JP	N/A
I404873	12/2012	TW	N/A
201825803	12/2017	TW	N/A
WO-2013051422	12/2012	WO	F16C 19/361

Primary Examiner: Johnson; Phillip A

Attorney, Agent or Firm: KIRTON McCONKIE

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/424,395 filed on Nov. 10, 2022, and entitled “REDUCER AND BEARING STRUCTURE THEREOF”. This application also claims priority to China Patent Application No. 202310712192.8, filed on Jun. 15, 2023. The entireties of the above-mentioned patent application are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION
(1) The present disclosure relates to a rotary bearing assembly, and more particularly to a rotary bearing assembly for providing the high load capacity and the high-speed ratio deceleration output, and having the advantage of small size.
BACKGROUND OF THE INVENTION
(2) Generally, a conventional rotary bearing includes a bearing inner ring, a bearing outer ring and a load element disposed between the bearing inner ring and the bearing outer ring. Through the rolling of the load element, the bearing outer ring being fixed and the bearing inner ring being rotating, or the bearing outer ring being rotating and the bearing inner ring being fixed can be achieved for certain applications. In general automation applications, if a load is to be carried and a rotary motion is to be performed, a rotary bearing will be used.
(3) On the other hand, the motor in automation applications operates at a high speed and a low torque, so it is difficult to use the motor to drive a large-sized load. For allowing the motor to drive a heavy object, a speed reducer is used for reducing the rotating speed of the motor, thereby increasing the torque. Therefore, the conventional rotary bearing is used in a combination of the motor and the reducer, and it is further combined with a gear set for torque output use. However, since the size of the rotary bearing varies as the size of the gear set mated therewith, it is unfeasible to use the same material in different applications. Therefore, when the rotary bearing is used with the gear set, a reserved space needs to be larger, which is unfavorable for miniaturization design.
(4) In addition, there are several different designs of reducers for performance improvement. However, it is not easy to achieve the purposes of reducing the entire volume, reducing the number of parts and facilitating the assembly under the same load at the same time. For example, in a conventional reducer, there are a plurality of pins placed between the inner teeth and the outer teeth, and the bearing ring and the inner gear ring are designed separately, so as to facilitate the axial fixation of the pins and the convenience of assembly. However, when the bearing ring and the inner gear ring are designed separately, the number of parts is increased, and the processing cost and assembly cost are increased. Furthermore, when the number of design parts is increased, the space requirement is increased, and it is difficult to achieve the volume miniaturization.
(5) In view of this, there is a need of providing a rotary bearing assembly for providing the high load capacity and the high-speed ratio deceleration output, and having the advantage of small size, so as to obviate the drawbacks encountered by the prior arts.

SUMMARY OF THE INVENTION

(6) An object of the present disclosure is to provide a rotary bearing assembly for forming an integrated bearing structure with the high load capacity and the high-speed ratio deceleration output, and also having the advantage of small size.

(7) Another object of the present disclosure is to provide a rotary bearing assembly. With the rotary bearing assembly combined with the reducer, it allows reducing the overall volume and the number of parts under the same load, facilitating to the assembly process, and solving the shortcomings of the conventional cycloidal reducer that are not conducive to miniaturization and cannot share bearing materials. On the other hand, the concentric end and the eccentric end of the transmission shaft used in the rotary bearing assembly have the same diameter. Consequently, it is not necessary to increase the diameter of the eccentric end, and the miniaturization of the cycloid speed reducer can be realized. In addition, since the concentric end and the eccentric end of the transmission shaft have the same diameter, the bearings disposed around the concentric end and the eccentric end of the transmission shaft respectively may be designed to be compatible with the same specification. In this way, the component cost is reduced. Moreover, the concentric end and the eccentric end of the transmission shaft are integrally formed as a one-piece structure. Consequently, the accuracy of the arrangement and alignment of the transmission shaft within the rotary bearing assembly is improved.

(8) In accordance with an aspect of the present disclosure, a rotary bearing assembly is provided and includes an input shaft, an inner-ring component, an outer-ring component and a load element. The input shaft is configured to combine a rotating shaft of a motor to provide a power input. The inner-ring component includes a gear set, wherein the inner-ring component is sleeved on the input shaft through the gear set and driven by the input shaft. The outer-ring component is sleeved on the inner-ring component through a load element and engaged with the gear set, wherein when the gear set is driven by the input shaft to drive the inner-ring component, the gear set drives the outer-ring component, and the inner-ring component and the outer-ring component are rotated relatively, wherein one of the inner-ring component and the outer-ring component is served to provide a power output, and a rotational speed difference is between the power input and the power output.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The Above Contents of the Present Disclosure Will Become More Readily Apparent to Those Ordinarily Skilled in the Art after Reviewing the Following Detailed Description and Accompanying Drawings, in which:

(2) FIG. **1** is a schematic structural view illustrating a rotary bearing assembly according to a first embodiment of the present disclosure;

(3) FIG. **2** is a vertically cross-sectional view illustrating the rotary bearing assembly combined with a motor according to the first embodiment of the present disclosure;

(4) FIG. **3** is a vertically cross-sectional view illustrating the rotary bearing assembly according to the first embodiment of the present disclosure;

(5) FIG. **4** is a horizontally cross-sectional view illustrating the rotary bearing assembly according to the first embodiment of the present disclosure;

(6) FIG. **5** is a lateral view illustrating the transmission shaft of the rotary bearing assembly according to the first embodiment of the present disclosure;

(7) FIG. **6** is a horizontally cross-sectional view illustrating a part of the rotary bearing assembly according to the first embodiment of the present disclosure;

(8) FIG. **7** is an enlarged view of region P**1** in FIG. **6**;

(9) FIG. **8** is an enlarged view of region P**2** in FIG. **7**;

(10) FIG. **9** is a vertically cross-sectional view illustrating a rotary bearing assembly according to a second embodiment of the present disclosure; and
(11) FIG. **10** is a lateral view illustrating the transmission shaft of the rotary bearing assembly according to the second embodiment of the present disclosure.
(12) FIG. **11** is a schematic exterior view illustrating a rotary bearing assembly according to a third embodiment of the present disclosure;
(13) FIG. **12** is a vertically cross-sectional view illustrating the rotary bearing assembly according to the third embodiment of the present disclosure;
(14) FIG. **13** is a schematic exploded view illustrating the rotary bearing assembly according to the third embodiment of the present disclosure;
(15) FIG. **14** is a schematic exploded view illustrating the rotary bearing assembly according to the third embodiment of the present disclosure and taken from another perspective;
(16) FIG. **15** is a top view illustrating the inner gear ring of the rotary bearing assembly according to the first embodiment of the present disclosure;
(17) FIG. **16** is a vertically cross-sectional view illustrating the inner gear ring of the rotary bearing assembly according to the first embodiment of the present disclosure;
(18) FIG. **17** is a horizontally cross-sectional view illustrating the rotary bearing assembly according to the third embodiment of the present disclosure;
(19) FIG. **18** is an enlarged view of region P**3** in FIG. **12**;
(20) FIG. **19** is a schematic exterior view illustrating a rotary bearing assembly according to a fourth embodiment of the present disclosure;
(21) FIG. **20** is a vertically cross-sectional view illustrating the rotary bearing assembly according to the fourth embodiment of the present disclosure;
(22) FIG. **21** is a schematic exploded view illustrating the outer-ring component of the rotary bearing assembly according to the fourth embodiment of the present disclosure; and
(23) FIG. **22** is a vertically cross-sectional view illustrating the outer-ring component of the rotary bearing assembly according to the fourth embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(24) The present disclosure will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as “top,” “bottom,” “upper,” “lower” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. When an element is referred to as being “connected,” or “coupled,” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. Although the wide numerical ranges and parameters of the present disclosure are approximations, numerical values are set forth in the specific examples as precisely as possible. In addition, although the “first,” “second,” “third,” and the like terms in the claims be used to describe the various elements can be appreciated, these elements should not be limited by

these terms, and these elements are described in the respective embodiments are used to express the different reference numerals, these terms are only used to distinguish one element from another element. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of example embodiments.

(25) Please refer to FIG. **1** and FIG. **2**. They illustrate a rotary bearing assembly according to a first embodiment of the present disclosure. Preferably but not exclusively, the rotary bearing assembly **1** is applied to various motor devices, machine tools, robotic arms, automobiles, motorcycles or other power machinery, so as to provide a power output with an appropriate speed difference.

(26) In the embodiment, a rotary bearing assembly **1** is provided and includes an input shaft **10**, an inner-ring component **20**, an outer-ring component **30** and a load element **40**. The input shaft **10** is located at a substantially central position of the rotary bearing assembly **1**, and configured to combine a rotating shaft **92** of a motor **90** to provide a power input. The inner-ring component **20** includes a gear set **21**. Preferably but not exclusively, the inner-ring component **20** is sleeved on the input shaft **10** through the gear set **21** and driven by the input shaft **10**. The outer-ring component **30** is sleeved on the inner-ring component **20** through the load element **40** and engaged with the gear set **21**. In the embodiment, when the gear set **21** is driven by the input shaft **10** to drive the inner-ring component **20**, the gear set **21** drives the outer-ring component **30**, and the inner-ring component **20** and the outer-ring component **30** are rotated relatively. In other words, one of the inner-ring component **20** and the outer-ring component **30** is served to provide a power output, and a rotational speed difference is between the power input and the power output.

(27) In the embodiment, the outer-ring component **30** is served as an output end, and the inner-ring component **20** is served as an input end. The inner-ring component **20** further includes at least one output disc **23**, **23′** and a transmission shaft **22**. Preferably but not exclusively, in the embodiment, the output disc **23** is fixed to an outer housing **91** of the motor **90**, and the output disc **23′** is fixed to a plane **8**. In the embodiment, the at least one output disc **23**, **23′** is linked with the gear set **21** through the transmission shaft **22**. When the gear set **21** is driven by the input shaft **10**, the out-ring component **30** is driven through the gear set **21** to provide the power output.

(28) Please refer to FIG. **3** together. Preferably but not exclusively, in another embodiment, the outer-ring component **30** is served as a fixed end, and the inner-ring component **20** is served as an output end. The inner-ring component **20** includes at least one output disc **23**, **23′** and a transmission shaft **22**. Preferably but not exclusively, in the embodiment, the at least one output disc **23**, **23′** is linked with the gear set **21** through the transmission shaft **22**. When the gear set **21** is driven by the input shaft **10**, the at last one output disc **23**, **23′** is driven through the transmission shaft **22** by the gear set **21** to provide the power output.

(29) In the embodiment, the outer-ring component **30** includes a needle-housing ring **300** and a plurality of rolling pins **301**. The plurality of rolling pins **301** are arranged on the needle-housing ring **300**. The load element **40** includes a plurality of bearing rollers **42** and a pair of track rings **43**, **44**. The pair of track rings **43**, **44** are disposed on two opposite sides of the needle-housing ring **300**, so as to provide a running track respectively for the plurality of bearing rollers **42** running thereon. In the embodiment, there are two running tracks **35** arranged in pairs, but the present disclosure is not limited thereto.

(30) In the embodiment, the gear set **21** includes a first cycloid disc **21***a* and a second cycloid disc **21***b*. Preferably but not exclusively, the first cycloid disc **21***a* is disposed around the input shaft **10** and driven by the input shaft **10** to rotate. The first cycloid disc **21***a* includes a first tooth part **210** contacted with at least corresponding one of the plurality of rolling pins **301**. Preferably but not exclusively, the second cycloid disc **21***b* is disposed around the input shaft **10** and driven by the input shaft **10** to rotate. The second cycloid disc **21***b* includes a second tooth part **210′** contacted with at least corresponding one of the rolling pins **301**. In the embodiment, the first cycloid disc **21***a* and the second cycloid disc **21***b* are located at two opposite sides of the needle-housing ring

300. In the embodiment, the pair of track rings **43**, **44** have a track inner diameter d**4**, the first cycloid disc **21***a* and the second cycloid disc **21***b* have a cycloid-disc outer diameter d**5**, and the track inner diameter d**4** is greater than the cycloid-disc outer diameter d**5**.

(31) In the embodiment, the at least one output disc **23**, **23′** includes a first output disc **23** and a second output disc **23′**. Preferably but not exclusively, the first output disc **23** and the second output disc **23′** are located at two opposite sides of the needle-housing ring **30**. In that, the first cycloid disc **21***a* and the second cycloid disc **21***b* are located between the first output disc **23** and the second output disc **23′**. Preferably but not exclusively, at least one of the first output disc **23** and the second output disc **23′** is served as the output end of the rotary bearing assembly **1** to provide the power output. Certainly, in another embodiment, the needle-housing ring **300** is served as the output end, and at least one of the first output disc **23** and the second output disc **23′** is served as the fixed end. The present disclosure is not limited thereto.

(32) Please refer to FIG. **3**, FIG. **4** and FIG. **5**. In the embodiment, the number of the transmission shafts **22** may be one or more. Preferably but not exclusively, there are five transmission shafts **22**. Each transmission shaft **22** is a crankshaft, and linked with the first cycloid disc **21***a*, the second cycloid disc **21***b*, the first output disc **23** and the second output disc **23′**. Preferably but not exclusively, each transmission shaft **22** includes a first concentric end **221**, a first eccentric end **222**, a second eccentric end **223** and a second concentric end **224** integrally formed as a one-piece structure and arranged sequentially. The first concentric end **221** is linked with the first output disc **23**, the first eccentric end **222** is linked with the first cycloid disc **21***a*, the second eccentric end **223** is linked with the second cycloid disc **21***b*, and the second concentric end **224** is linked with the second output disc **23′**. Moreover, in the embodiment, an eccentricity value is between any neighboring two of the first concentric end **221**, the first eccentric end **222**, the second eccentric end **223** and the second concentric end **224**, and a diameter ΦA of the first concentric end **221**, a diameter ΦC of the first eccentric end **222**, a diameter ΦD of the second eccentric end **223** and a diameter ΦB of the second concentric end **224** are all equal to a transmission-shaft diameter d**1** of the transmission shaft **22**.

(33) From the above descriptions, instead of limiting the diameter of the eccentric end of the transmission shaft **22** to be greater than the diameter of the concentric end of the transmission shaft **22**, the rotary bearing assembly **1** in the embodiment of the present disclosure allows the first concentric end **221**, the first eccentric end **222**, the second eccentric end **223** and the second concentric end **224** of the transmission shaft **22** to have the same diameter (i.e., ΦA=ΦC=ΦD=ΦB), which is equal to the transmission-shaft diameter d**1**. Consequently, it is not necessary to increase the diameter of the first eccentric end **222** and the diameter of the second eccentric end **223**, and the miniaturization of the rotary bearing assembly **1** can be realized. As mentioned above, the first concentric end **221**, the first eccentric end **222**, the in second eccentric end **223** and the second concentric end **224** of the transmission shaft **22** have the same diameter. Consequently, when the bearings are disposed around the first concentric end **221**, the first eccentric end **222**, the second eccentric end **223** and the second concentric end **224** of the transmission shaft **22** respectively, these bearings may be designed to be compatible with the same specification. In this way, the component cost is reduced. Moreover, the first concentric end **221**, the first eccentric end **222**, the second eccentric end **223** and the second concentric end **224** of the transmission shaft **22** are integrally formed as a one-piece structure. Consequently, the accuracy of the arrangement and alignment of the transmission shaft **22** within the rotary bearing assembly **1** is improved.

(34) In the embodiment, as shown in FIG. **5**, the eccentricity value between the first concentric end **221** and the first eccentric end **222** is defined as a first eccentricity value E**1**. The eccentricity value between the second concentric end **224** and the second eccentric end **223** is defined as a second eccentricity value E**2**. The eccentricity value between the first eccentric end **222** and the second eccentric end **223** is defined as a third eccentricity value E**3**. The first eccentricity value E**1** is equal to the second eccentricity value E**2**, and the third eccentricity value E**3** is double the first

eccentricity value E**1**. Accordingly, it is conducive to dispose the bearings around the first concentric end **221**, the first eccentric end **222**, the second eccentric end **223** and the second concentric end **224** of the transmission shaft **22**, respectively. In addition, in the embodiment, the first concentric end **221** and the second concentric end **224** are coaxial with the transmission shaft **22**. The first eccentric end **222** and the second eccentric end **223** are eccentrically disposed on the transmission shaft **22**. The eccentric direction of the first eccentric end **222** is opposite to the eccentric direction of the second eccentric end **223**.

(35) Please also refer to FIG. **6**. In some embodiments, the first cycloid disc **21***a* further includes at least one installation hole **211**. Each installation hole **221** is aligned with the corresponding transmission shaft **22** for allowing the first eccentric end **222** of the corresponding transmission shaft **22** to penetrate through. Consequently, the first eccentric end **222** of the corresponding transmission shaft **22** and the first cycloid disc **21***a* are linked with each other. In the embodiment, the first output disc **23** includes at least one installation hole **230**. Each installation hole **230** is aligned with the corresponding transmission shaft **22** for allowing the first concentric end **221** of the corresponding transmission shaft **22** to penetrate through. Consequently, the first concentric end **221** of the corresponding transmission shaft **22** and the first output disc **23** are linked with each other. In the embodiment, the second cycloid disc **21***b* further includes at least one installation hole **211′**. Each installation hole **211′** is aligned with the corresponding transmission shaft **22** for allowing the second eccentric end **223** of the corresponding transmission shaft **22** to penetrate through. Consequently, the second eccentric end **223** of the corresponding transmission shaft **22** and the second cycloid disc **21***b* are linked with each other. In the embodiment, the second output disc **23′** includes at least one installation hole **230′**. Each installation hole **230′** is aligned with the corresponding transmission shaft **22** for allowing the second concentric end **224** of the corresponding transmission shaft **22** to penetrate through. Consequently, the second concentric end **224** of the corresponding transmission shaft **22** and the second output disc **23′** are linked with each other.

(36) In addition, in the embodiment, the rotary bearing assembly **1** further includes a first bearing **24** and a second bearing **24′**. The structure of the first bearing **24** and the structure of the second bearing **24′** are the same. The first bearing **24** is disposed between the installation hole **211** of the first cycloid disc **21***a* and the first eccentric end **222**. The second bearing **24′** is disposed between the installation hole **211′** of the second cycloid disc **21***b* and the second eccentric end **223**. Moreover, each of the first bearing **24** and the second bearing **24′** includes a plurality of output eccentric shaft needles **241**. Since the first bearing **24** and the second bearing **24′** have the same structure, only the output eccentric shaft needles **241** of the first bearing **24** are exemplified in FIG. **6**. The plurality of output eccentric shaft needles **241** of the first bearing **24** are disposed around the main body of the first bearing **24**. When the first bearing **24** is disposed between the installation hole **211** of the first cycloid disc **21***a* and the first eccentric end **222**, the plurality of output eccentric shaft needles **241** of the first bearing **24** are disposed around the outer ring wall of the first eccentric end **222**. Similarly, the plurality of output eccentric shaft needles **241** of the second bearing **24′** are disposed around the main body of the second bearing **24′**. When the second bearing **24′** is disposed between the installation hole **211′** of the second cycloid disc **21***b* and the second eccentric end **223**, the plurality of output eccentric shaft needles **241** of the second bearing **24′** are disposed around the outer ring wall of the second eccentric end **223**. In addition, the transmission-shaft diameter d**1** of the transmission shaft **22** is equal to the diameter ΦA of the first concentric end **221**, the diameter ΦC of the first eccentric end **222**, the diameter ΦD of the second eccentric end **223** and the diameter ΦB of the second concentric end **224** are d**1** (i.e., ΦA=ΦC=ΦD=ΦB). Please further refer to FIG. **7**. The diameter of each output eccentric shaft needle **241** is d**2**. The diameter of the installation hole **211** and the diameter of the installation hole **211′** are both d**3**. The diameter d**3** of the installation hole **211** and the installation hole **211′** is equal to a sum of the diameter d**1** of the first eccentric end **222** and double of the diameter d**2** of the output eccentric

shaft needle **2411** (i.e., d**3**=d**1**+2d**2**). Moreover, double of the diameter d**2** of the output eccentric shaft needle **241** is greater than or equal to the first eccentricity value E**1** (i.e., 2d≥E**1**).

(37) Please refer to FIG. **3**, FIG. **5**, FIG. **7** and FIG. **8**. In the embodiment, each of the first bearing **24** and the second bearing **24′** further includes a plurality of rolling pin retainers **242**, each adjacent two of the plurality of output eccentric shaft needles **241** are spaced by one rolling pin retainer **242**, which is configured for covering and supporting the output eccentric shaft needle **241**. In addition, a gap G is formed between each of the plurality of output eccentric shaft needles **241** and the adjacent corresponding rolling pin retainer **242** within the corresponding bearing. Therefore, under the circumstance that double of the diameter d**2** of the output eccentric shaft needle **241** is greater than or equal to the first eccentricity value E**1**, the first bearing **24** and the second bearing **24′** are ensured to be sleeved on the first eccentric end **222** and the second eccentric end **223**, respectively. In addition, the link the first concentric end **221** and the first output disc **23**, and the link between the second concentric end **224** and the second output disc **23′** are implemented through the bearings, which are similar to the first bearing **24** and the second bearing **24′** in structure and size. The present disclosure is not limited thereto, and not redundantly described herein.

(38) Please refer to FIG. **1**, FIG. **3** and FIG. **5**. In the embodiment, the first cycloid disc **21***a* and the second cycloid disc **21***b* are disposed within the needle-housing ring **300**. Furthermore, the at least one output disc **23**, **23′** of the inner-ring component **20** or the needle-housing ring **300** of the outer-ring component **30** can be driven to rotate by the first cycloid disc **21***a* and the second cycloid disc **21***b*. In an embodiment, the power transmission mode of the rotary bearing assembly **1** is described as the following. When the input shaft **10** is rotated, the first cycloidal disc **21***a* and the second cycloidal disc **21***b* are driven by the input shaft **10** to rotate. In addition, the first cycloidal disc **21***a* and the second cycloidal disc **21***b* linked with the first eccentric end **222** and the second eccentric end **223** of the transmission shaft **22**, respectively, further drive the rotation of the transmission shaft **22**. Consequently, the first concentric end **221** and the second concentric end **224** of the transmission shaft **22** are synchronously rotated and respectively drive the rotations of the first output disc **23** and the second output disc **23′**. That is, the first output disc **23** and/or the second output disc **23′** is served to provide the power output of the rotary bearing assembly **1**. In an embodiment, the first output disc **23** and/or the second output disc **23′** is fixed, and the needle-housing ring **300** is served to provide the power output of the rotary bearing assembly **1**. The present disclosure is not limited thereto, and not redundantly described herein.

(39) Please refer to FIG. **1**, FIG. **9** and FIG. **10**. A rotary bearing assembly is provided according to a second embodiment of the present disclosure. In the embodiment, the structures, elements and functions of the rotary bearing assembly **1***a* are similar to those of the rotary bearing assembly **1** of FIG. **1** to FIG. **8**, and are not redundantly described herein. Unlike the rotary bearing assembly **1** shown in FIG. **1** to FIG. **8** including two cycloid discs, the rotary bearing assembly **1***a* in this embodiment includes one single cycloid disc. That is, the rotary bearing assembly **1***a* includes a first cycloid disc **21***a*. The first cycloid disc **21***a* is disposed around the input shaft **10** and driven by the input shaft **10** to rotate, and the first cycloid disc **21***a* includes a first tooth part **210** contacted with at least corresponding one of the plurality of rolling pins **301**. Preferably but not exclusively, a power input provided by a motor (not shown) is received by the input shaft **10**, and the input shaft **10** is driven by the power input **10** to rotate. The input shaft **10** is located at a substantially central position of the rotary bearing assembly **1***a*.

(40) In the embodiment, the at least one output disc **23**, **23′** includes a first output disc **23** and a second output disc **23′** as shown in FIG. **9**. Preferably but not exclusively, the first output disc **23** and the second output disc **23′** are located at two opposite sides of the needle-housing ring **30**. In that, the first cycloid disc **21***a* is located between the first output disc **23** and the second output disc **23′**. Preferably but not exclusively, at least one of the first output disc **23** and the second output disc **23′** is served as the output end of the rotary bearing assembly **1***a* to provide the power output. Certainly, in another embodiment, the first output disc **23** and the second output disc **23′** are served

as a fixed end, and the outer-ring component **30** is served as the output end. The present disclosure is not limited thereto.
(41) In the embodiment, the transmission shaft **22** is a crankshaft, and linked with the first cycloid disc **21***a*, the first output disc **23** and the second output disc **23**′. Preferably but not exclusively, the transmission shaft **22***a* includes a first concentric end **221***a*, a first eccentric end **222***a* and a second concentric end **223***a* integrally formed as a one-piece structure and arranged sequentially. The first concentric end **221***a* is linked with the first output disc **23**, the first eccentric end **222***a* is linked with the first cycloid disc **21***a*, and the second concentric end **223***a* is linked with the second output disc **23**′. Moreover, in the embodiment, an eccentricity value is between any neighboring two of the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a*, and a diameter ΦA of the first concentric end **221***a*, a diameter **1**C of the first eccentric end **222***a* and a diameter ΦB of the second concentric end **223***a* are all equal to a transmission-shaft diameter d**1** (as shown in FIG. **7**) of the transmission shaft **22***a*.
(42) In the embodiment, instead of limiting the diameter of the eccentric end of the transmission shaft **22***a* to be greater than the diameter of the concentric end of the transmission shaft **22***a*, the rotary bearing assembly **1***a* in the embodiment of the present disclosure allows the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a* of the transmission shaft **22***a* to have the same diameter (i.e., ΦA=ΦC=ΦB), which is equal to the transmission-shaft diameter d**1**. Consequently, it is not necessary to increase the diameter of the first eccentric end **222***a*, and the miniaturization of the rotary bearing assembly **1***a* can be realized. As mentioned above, the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a* of the transmission shaft **22***a* have the same diameter. Consequently, when the bearings are disposed around the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a* of the transmission shaft **22***a* respectively, these bearings may be designed to be compatible with the same specification. In this way, the component cost is reduced. Moreover, the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a* of the transmission shaft **22***a* are integrally formed as a one-piece structure. Consequently, the accuracy of the arrangement and alignment of the transmission shaft **22***a* within the rotary bearing assembly **1***a* is improved.
(43) In the embodiment, as shown in FIG. **10**, the eccentricity value between the first concentric end **221***a* and the first eccentric end **222***a* is defined as a first eccentricity value E**1**. The eccentricity value between the second concentric end **223***a* and the first eccentric end **222***a* is defined as a second eccentricity value E**2**. The first eccentricity value E**1** is equal to the second eccentricity value E**2**. Accordingly, it is conducive to dispose the bearings around the first concentric end **221***a*, the first eccentric end **222***a* and the second concentric end **223***a* of the transmission shaft **22***a*, respectively. In addition, in the embodiment, the first concentric end **221***a* and the second concentric end **223***a* are coaxial with the transmission shaft **22***a*. The first eccentric end **222***a* is eccentrically disposed on the transmission shaft **22***a*.
(44) Similarly, in an embodiment, the power transmission mode of the rotary bearing assembly **1***a* is described as the following. When the input shaft **10** is rotated, the first cycloidal disc **21***a* is driven by the input shaft **10** to rotate. In addition, the first cycloidal disc **21***a* linked with the first eccentric end **222***a* of the transmission shaft **22***a*, respectively, further drives the rotation of the transmission shaft **22***a*. Consequently, the first concentric end **221***a* and the second concentric end **223***a* of the transmission shaft **22***a* are synchronously rotated and respectively drive the rotations of the first output disc **23** and the second output disc **23**′. That is, the first output disc **23** and/or the second output disc **23**′ is served to provide the power output of the rotary bearing assembly **1***a*. In an embodiment, the first output disc **23** and/or the second output disc **23**′ is fixed, and the outer-ring component **30** is served to provide the power output of the rotary bearing assembly **1***a*. The present disclosure is not limited thereto, and not redundantly described herein.
(45) Please refer to FIG. **11** to FIG. **18**. A rotary bearing assembly is provided according to a third

embodiment of the present disclosure. Please refer to FIGS. **11** to **13**. In the embodiment, the structures, elements and functions of the rotary bearing assembly **1***b* are similar to those of the rotary bearing assembly **1** of FIG. **1** to FIG. **8**, and are not redundantly described herein. In the embodiment, the load element **40***a* includes a plurality of bearing rollers **42**. Preferably but not exclusively, the outer-ring component **30** is an inner gear ring, which includes an inner-gear-ring main body **31**, an inner tooth portion **32** and at least one running track **35** for the plurality of bearing rollers **42** running thereon. In the embodiment, there are two running tracks arranged in pairs. Moreover, the running track **35** has an inclined angle relative to an axial direction of the input shaft **10**. In the embodiment, the gear set **21** of the inner-ring component **20** and the inner tooth portion **32** of the inner gear ring are partially engaged with each other.

(46) Please refer to FIG. **14** to FIG. **15**. In the embodiment, the inner tooth portion **32** is set on an inner-base surface **33** of the inner-gear-ring main body **31** and has a gear-base diameter D**1**. The running track **35** is arranged on one lateral side of the inner gear ring has a track inner diameter D**2** in the innermost circle **34** adjacent to the inner tooth portion **32**. Preferably but not exclusively, the track inner diameter D**2** is greater than or equal to the gear-base diameter D**1**. In the embodiment, the running track **35** is spatially corresponding to the inner-gear-ring main body **31** and the at least one output disc **23′** of the inner-ring component **20**, and a parallelogram is collaboratively formed on a radial section so that the plurality of bearing rollers **42** are configured to run between the inner-gear-ring main body **31** and the at least one output disc **23′** of the inner-ring component **20**.

(47) Please refer to FIG. **14** to FIG. **17**. In the embodiment, when the input shaft **10** is rotated, the first cycloidal disc **21***a* and the second cycloidal disc **21***b* are driven by the input shaft **10** to rotate. In addition, the first cycloidal disc **21***a* and the second cycloidal disc **21***b* linked with the transmission shaft **22** further drive the rotation of the transmission shaft **22**. Consequently, the first output disc **23** and the second output disc **23′** are driven through the transmission shaft **22** to rotate. In the embodiment, the first output disc **23** and/or the second output disc **23′** is served to provide the power output of the rotary bearing assembly **1***b*. Preferably but not exclusively, the second output disc **23′** is served to provide the power output, and the outside of the first output disc **23** is covered with a fixed housing **80**. The fixed housing **80** is combined with the outer-ring component **30***a* and served as a fixed end. Certainly, in another embodiment, the first output disc **23** and/or the second output disc **23′** is fixed, and the outer-ring component **30***a* is served to provide the power output of the rotary bearing assembly **1***b*. The present disclosure is not limited thereto, and not redundantly described herein.

(48) Please refer to FIG. **19** to FIG. **20**. A rotary bearing assembly is provided according to a fourth embodiment of the present disclosure. In the embodiment, the structures, elements and functions of the rotary bearing assembly **1***c* are similar to those of the rotary bearing assembly **1** of FIG. **1** to FIG. **8**, and are not redundantly described herein. In the embodiment, the outer-ring component **30***b* includes a pair of roller retaining rings **39** and a running track **35**. The pair roller retaining rings **39** are disposed on the two opposite sides of the outer-ring component **30***b* through an accommodation slot **36**, respectively. The running tracks **35** are located at the two opposite sides of the outer-ring component **30***b*, and disposed adjacent to the roller retaining rings **39**. In the embodiment, the load element **40***b* includes a plurality of bearing rollers **42**, which are configured to run on the running tracks **35**.

(49) In summary, the present disclosure provides a rotary bearing assembly for forming an integrated bearing structure with the high load capacity and the high-speed ratio deceleration output, and also having the advantage of small size. With the rotary bearing assembly combined with the reducer, it allows reducing the overall volume and the number of parts under the same load, facilitating to the assembly process, and solving the shortcomings of the conventional cycloidal reducer that are not conducive to miniaturization and cannot share bearing materials. On the other hand, the concentric end and the eccentric end of the transmission shaft used in the rotary bearing assembly have the same diameter. Consequently, it is not necessary to increase the diameter

of the eccentric end, and the miniaturization of the cycloid speed reducer can be realized. In addition, since the concentric end and the eccentric end of the transmission shaft have the same diameter, the bearings disposed around the concentric end and the eccentric end of the transmission shaft respectively may be designed to be compatible with the same specification. In this way, the component cost is reduced. Moreover, the concentric end and the eccentric end of the transmission shaft are integrally formed as a one-piece structure. Consequently, the accuracy of the arrangement and alignment of the transmission shaft within the rotary bearing assembly is improved.

(50) While the disclosure has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the disclosure needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A rotary bearing assembly comprising: an input shaft configured to combine a rotating shaft of a motor to provide a power input; an inner-ring component comprising a gear set, wherein the inner-ring component is sleeved on the input shaft through the gear set and driven by the input shaft; and an outer-ring component sleeved on the inner-ring component through a load element and engaged with the gear set, wherein when the gear set is driven by the input shaft to drive the inner-ring component, the gear set drives the outer-ring component, and the inner-ring component and the outer-ring component are rotated relatively, wherein one of the inner-ring component and the outer-ring component is served to provide a power output, and a rotational speed difference is between the power input and the power output, wherein the load element includes a pair of track rings, the gear set comprises a first cycloid disc, and the first cycloid disc is disposed around the input shaft and driven by the input shaft to rotate, wherein the pair of track rings have a track inner diameter, the first cycloid disc has a cycloid-disc outer diameter, and the track inner diameter is greater than the cycloid-disc outer diameter.

2. The rotary bearing assembly according to claim 1, wherein the inner-ring component comprises at least one output disc and a transmission shaft, and the at least one output disc is linked with the gear set through the transmission shaft.

3. The rotary bearing assembly according to claim 2, wherein in case of that the outer-ring component is served as an output end and the inner-ring component is served as a fixed end, the at least one output disc is fixed, the gear set is driven by the input shaft, and the outer-ring component is driven through the gear set to provide the power output, wherein in case of that the outer-ring component is served as a fixed and the inner-ring component is served an output end, the outer-ring component is fixed, the gear set is driven by the input shaft, and the at last one output disc is driven through the transmission shaft by the gear set to provide the power output.

4. The rotary bearing assembly according to claim 2, wherein the outer-ring component comprises a needle-housing ring and a plurality of rolling pins, the plurality of rolling pins are arranged on the needle-housing ring, the load element includes a plurality of bearing rollers, and the pair of track rings are disposed on two opposite sides of the needle-housing ring, so as to provide a running track respectively for the plurality of bearing rollers running thereon.

5. The rotary bearing assembly according to claim 4, wherein the gear set further comprises a second cycloid disc, the first cycloid disc comprises a first tooth part contacted with at least corresponding one of the plurality of rolling pins, the second cycloid disc is disposed around the input shaft and driven by the input shaft to rotate, and the second cycloid disc comprises a second tooth part contacted with at least corresponding another one of the rolling pins, wherein the first cycloid disc and the second cycloid disc are located at two opposite sides of the needle-housing

ring.
6. The rotary bearing assembly according to claim 5, wherein the second cycloid disc has the cycloid-disc outer diameter.
7. The rotary bearing assembly according to claim 5, wherein the transmission shaft comprises a first concentric end, a first eccentric end, a second eccentric end and a second concentric end integrally formed as a one-piece structure and arranged sequentially, wherein the first eccentric end is linked with the first cycloid disc, the second eccentric end is linked with the second cycloid disc, an eccentricity value is between any neighboring two of the first concentric end, the first eccentric end, the second eccentric end and the second concentric end, and a diameter of the first concentric end, a diameter of the first eccentric end, a diameter of the second eccentric end and a diameter of the second concentric end are all equal to a transmission-shaft diameter, wherein the at least one output disc is linked with the first concentric end or the second concentric end.
8. The rotary bearing assembly according to claim 7, wherein the eccentricity value between the first concentric end and the first eccentric end is defined as a first eccentricity value, the eccentricity value between the second concentric end and the second eccentric end is defined as a second eccentricity value, the eccentricity value between the first eccentric end and the second eccentric end is defined as a third eccentricity value, the first eccentricity value is equal to the second eccentricity value, and the third eccentricity value is double the first eccentricity value.
9. The rotary bearing assembly according to claim 8, wherein the inner-ring component further comprises a first bearing and a second bearing, each of the first bearing and the second bearing comprises a plurality of output eccentric shaft needles, and double of a diameter of each of the plurality of output eccentric shaft needles is greater than or equal to the first eccentricity value.
10. The rotary bearing assembly according to claim 9, wherein the plurality of output eccentric shaft needles of the first bearing are disposed around an outer ring wall of the first eccentric end, and the plurality of output eccentric shaft needles of the second bearing are disposed around an outer ring wall of the second eccentric end.
11. The rotary bearing assembly according to claim 10, wherein the at least one output disc comprises a first output disc and a second output disc located at two opposite sides of the needle-housing ring, the first output disc comprises an installation hole aligned with the corresponding transmission shaft for allowing the first concentric end of the corresponding transmission shaft to penetrate through, and the second output disc comprises an installation hole aligned with the corresponding transmission shaft for allowing the second concentric end of the corresponding transmission shaft to penetrate through.
12. The rotary bearing assembly according to claim 11, wherein each of the installation holes has an inner diameter equal to a sum of the transmission-shaft diameter and double of a diameter of the plurality of output eccentric shaft needles.
13. The rotary bearing assembly according to claim 9, wherein each adjacent two of the plurality of output eccentric shaft needles are spaced by one corresponding rolling pin retainer, and a gap is formed between each of the plurality of output eccentric shaft needles and the corresponding rolling pin retainer.
14. The rotary bearing assembly according to claim 4, wherein the first cycloid disc comprises a first tooth part contacted with at least corresponding one of the plurality of rolling pins.
15. The rotary bearing assembly according to claim 14, wherein the transmission shaft comprises a first concentric end, a first eccentric end and a second concentric end integrally formed as a one-piece structure and arranged sequentially, wherein the first eccentric end is linked with the first cycloid disc, an eccentricity value is between any neighboring two of the first concentric end, the first eccentric end and the second concentric end, and a diameter of the first concentric end, a diameter of the first eccentric end and a diameter of the second concentric end are all equal to a transmission-shaft diameter, wherein the at least one output disc is linked with the first concentric end or the second concentric end.

16. The rotary bearing assembly according to claim 15, wherein the eccentricity value between the first concentric end and the first eccentric end is defined as a first eccentricity value, the eccentricity value between the second concentric end and the first eccentric end is defined as a second eccentricity value, and the first eccentricity value is equal to the second eccentricity value.
17. A rotary bearing assembly comprising: an input shaft configured to combine a rotating shaft of a motor to provide a power input; an inner-ring component comprising a gear set, wherein the inner-ring component is sleeved on the input shaft through the gear set and driven by the input shaft; and an outer-ring component sleeved on the inner-ring component through a load element and engaged with the gear set, wherein when the gear set is driven by the input shaft to drive the inner-ring component, the gear set drives the outer-ring component, and the inner-ring component and the outer-ring component are rotated relatively, wherein one of the inner-ring component and the outer-ring component is served to provide a power output, and a rotational speed difference is between the power input and the power output, wherein the inner-ring component comprises at least one output disc and a transmission shaft, and the at least one output disc is linked with the gear set through the transmission shaft, wherein the load element comprises a plurality of bearing rollers, and the outer-ring component includes an inner gear ring comprising an inner-gear-ring main body, an inner tooth portion and a running track for the plurality of bearing rollers running thereon, and the running track has an inclined angle relative to an axial direction of the input shaft, wherein the gear set of the inner-ring component and the inner tooth portion of the inner gear ring are partially engaged with each other.
18. The rotary bearing assembly according to claim 17, wherein the inner tooth portion is set on an inner-base surface of the inner-gear-ring main body and has a gear-base diameter, the running track is arranged on one lateral side of the inner gear ring has a track inner diameter adjacent to the inner tooth portion, wherein the track inner diameter is greater than or equal to the gear-base diameter.
19. The rotary bearing assembly according to claim 17, wherein the running track is spatially corresponding to the inner-gear-ring main body and the at least one output disc of the inner-ring component, and a parallelogram is collaboratively formed on a radial section so that the plurality of bearing rollers are configured to run between the inner-gear-ring main body and the at least one output disc of the inner-ring component.